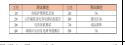
FACTOR TREE ANALYSIS FOR NON DETECTION - Why the problem was not detected ? [14]

PROBLEM DESCRIPTI OK件：刷盒尺寸使用通止规测试，通规通，止规止，外观OK，刷盒内腔无异物，刷握板面无松香残留，性能测试OK。NG件：无返修件标识，为正常批量件，刷盒尺寸使用通止规测试，通规通，止规止，外观OK，刷盒内腔有异物、异物成分分析为松香刷握板面无松香残

4M [2]	FACTORS [3]	STANDARD [5]		Factor VALUE on OK PARTS [6] Good Part / Situation taken as reference [8]	Factor VALUE on BAD PARTS [7] Bad Part / Situation taken as reference [9]	JUDGMENT Std OK [10]	Meet std [11]	Direct link [12]	Potential Root Causes	1) Ask WHY until you arrive at the real root cause. 2) For each WHY level, decide if a countermeasure (corrective or preventive action) is to be defined and put a 'Y'. 5 WHYs		Countermeasure?
METHOD	功能测试不合格品管理 Management of non-conforming products in functional testing	What was the standard value and tolerance?	收集功能不良，集中摆放，当天进行1#-8#工位电路测试，复测品在正面热保护器处打点标识 If the collection function is poor, place them in a centralized manner. Conduct circuit tests at stations 1# to 8# on the same day. Mark the retested items at the thermal protector on the front	正常下件产品为OK件部需复测，当功能测试不合格时，按复测流程进行（详细）工位复测，复测合格品打点，NG件报废 For normal delivered products, the OK parts need to be retested. When the functional test fails, the retest process should be followed (in detail) for the workstation retest. The retested qualified products should be marked, while the NG products should be scrapped. 	坏件产品正反面无复测打点标识，为正常下线OK件 Under normal circumstances, products that are OK parts need to be retested. When the functional test fails, the retest should be conducted at the workstations according to the detailed retest process. Products that pass the retest should be processed, and those that fail the retest should be scrapped	Y	Y	N	N/A	1st WHY		
		Where was the standard value defined (name, number and index of the document) ?	文件：误判品/返工品复测 ，文件编号：BC0100W1130 RDocument: Retesting of misjudged products/reworked products, Document Number: BC0100W1130R							2nd WHY		
		How was the standard value measured or verified?	性能通过电路测试机，外观通过CCD测试 The performance passed the circuit test machine, and the appearance passed the CCD test							3rd WHY		
		What was the frequency of measurement or verification?	1次/班Once per class							4th WHY		
										5th WHY		
MATERIAL	碳刷导通测试 Carbon brush conduction test	What was the standard value and tolerance?	碳刷导通测试，导柱顶出保持环，推杆反复推压碳刷三次，每次碳刷回弹与推杆接触，则显示导通。若一次碳刷未与推杆接触，则会系统报警 In the carbon brush conduction test, the guide column is pushed out of the holding ring, and the push rod repeatedly pushes the carbon brush three times. Each time the carbon brush rebounds and comes into contact with the push rod, it indicates conduction. If the carbon brush does not come into contact with the push rod at one time, the system will alarm	机器运转正常，无异常记录，开班Master Sample验证，机器可以正常识别碳刷卡滞的三件失效样品 The machine is operating normally with no abnormal records. The Master Sample verification at the start of the class shows that the machine can normally identify the three failed samples with stuck carbon swiping cards 	机器运转正常，无异常记录，开班Master Sample验证，机器可以正常识别碳刷卡滞的三件失效样品 The machine is operating normally with no abnormal records. The Master Sample verification at the start of the class shows that the machine can normally identify the three failed samples with stuck carbon swiping cards （NG件测试机） 刷盒内不同松香量在测试机的测试结果，	D	Y	D	碳刷回弹性能测试采用导通方式检测，未有具体力值监控，若碳刷存在轻微卡滞，回弹力若存在误判风险 The test for the rebound performance of the carbon brush is conducted using the conduction method. There is no specific force value monitoring. If there is slight sticking of the carbon brush, there is a risk of incorrect judgment regarding the rebound force.	1st WHY	碳刷存在轻微卡滞或者回弹力弱，机器无法识别If the carbon brush has slight sticking or lacks rebound force, the machine will not be able to recognize it.	
		Where was the standard value defined (name, number and index of the document) ?	文件：电路测试标准书，文件编号：BC0100W1130 Document: Circuit Test Standard Book, Document Number: BC0100W1130							2nd WHY	小，接触导通的缺陷 The testing machine can only identify defects such as misalignment, inability to conduct, low elasticity, and contact conductivity.	YES
		How was the standard value measured or verified?	弹力NG件，机器屏幕显示碳刷A/B/C断路， Elastic NG component, the machine screen shows that the carbon brush A/B/C is open-circuited.							3rd WHY	也性能测试机开发时未考虑该功能 This function was not taken into account during the development of the electrical performance testing machine.	
		What was the frequency of measurement or verification?	100%							4th WHY		
										5th WHY		
METHOD	终检与GP12外观检验 Final inspection and GP12 appearance inspection	What was the standard value and tolerance?	检验员依据外观检验WI对产品进行100%外观全检 The inspector conducts a 100% full appearance inspection of the product based on the appearance inspection WI	OK件依据外观检验WI，未发现有外观异常，判定OK品 According to the appearance inspection WI, no appearance abnormalities were found for the OK item, and it was determined to be an OK product	NG件依据外观检验WI，未发现有外观异常，碳刷滑动测试，碳刷可以正常复位和弹出，判定OK According to the visual inspection WI, no appearance abnormalities were found for NG parts. The carbon brush sliding test showed that the carbon brush could pop out normally, and it was judged OK	Y	Y	D	终检、GP12外观检验标准无法识别卡顿失效 The appearance inspection standard cannot identify stuttering failure	1st WHY	前面工序有回弹100%检测，GP12检验WI中检验产品外观，未做性能进行再次检测 The previous process had a 100% rebound test. The appearance of the product was inspected in the GP12 test WI, and no performance was tested again	
		Where was the standard value defined (name, number and index of the document) ?	文件：外观检验基准书，文件编号：BC018QW1150 Document: Appearance Inspection Reference Book, Document Number: BC018QW1150							2nd WHY	CP中未识别终检验及GP12过程中可能存在碳刷卡滞的异常风险 The CP did not identify the abnormal risk of carbon card lag during the final inspection and the GP12	YES
		How was the standard value measured or verified?	只进行产品外观全检，弹簧回弹性能未做检查 Only a full inspection of the product appearance was conducted, and the spring resilience performance was not checked							3rd WHY	FMEA中未识别潜在风险卡顿 Potential risks were not identified in the FMEA	
		What was the frequency of measurement or verification?	100%							4th WHY		
										5th WHY		
METHOD	锡焊后刷盒外观检验 Appearance inspection of brush box after soldering	What was the standard value and tolerance?	依据锡焊作业指导书进行作业，完成后对产品外观进行自检 Operate according to the soldering operation manual	OK品产品外观良好，无短路、虚焊，焊锡完全覆盖引脚，焊点数量正确，刷盒内部无助焊剂残留或氧化 The appearance is good, with no short circuit or false soldering. The solder completely covers the pins, and the number of solder joints is correct. There is no flux residue or oxidation inside the brush box.	NG品刷握板无松香残留，刷盒内口部有松香残留物 There is no residual rosin on the brush grip plate, and there is residual rosin at the mouth of the brush box	N	N	Y	工序作业自检，无100%刷盒内腔全检 Self inspection of process operation, no 100% full inspection of the inner cavity of the brush box	1st WHY	刷盒内有松香残留 There is residual rosin in the brush box	
		Where was the standard value defined (name, number and index of the document) ?	BC0100W1060手工锡焊作业指导书 BC0100W1060 Manual Soldering Operation Manual							2nd WHY	作业员未进行全面自检 The operator did not conduct a comprehensive self inspection	
		How was the standard value measured or verified?	目视 Visual							3rd WHY	锡焊作业指导书中未要求进行刷盒内腔自检 The soldering operation manual does not require self inspection of the inner cavity of the brush box	YES
		What was the frequency of measurement or verification?	每班 Per class							4th WHY	PFMEA中未识别刷盒中残留松香的风险 Risk of residual rosin in brush boxes not identified in PFMEA	
										5th WHY		